

Form Correctness Detection Using Pose Estimation

1. Introduction

This project presents a real-time form correctness detection system for physical exercises using pose estimation. The system analyzes human body movements from video input and provides repetition counting and corrective feedback to help users maintain proper exercise posture.

2. Objective

The objective of this project is to monitor exercise posture using body joint angles and predefined biomechanical rules. The system ensures correct form execution to reduce injury risk and improve workout effectiveness.

3. System Architecture

The application follows a modular pipeline that processes video input, extracts body keypoints, calculates joint angles, applies rule-based posture validation, and generates real-time feedback.

- Video Input: Live webcam feed or uploaded exercise video.
- Pose Estimation: MediaPipe Pose model detects 33 human landmarks.
- Angle Calculation: Joint angles computed using geometric relationships.
- Rule-Based Logic: Exercise-specific posture rules are evaluated.
- Feedback System: Corrective instructions are shown in real time.

4. Pose Estimation

MediaPipe Pose is used to extract body landmarks such as shoulders, elbows, wrists, and hips. This model operates efficiently in real time without the need for training.

5. Angle Calculation

Angles are calculated using three joint points to measure movement at critical joints. A moving average filter is applied to smooth angle values and reduce noise.

6. Supported Exercises

- Bicep Curl
- Lateral Raise

7. Rule-Based Posture Validation

Posture correctness is evaluated using deterministic rules derived from exercise biomechanics. Each frame is independently analyzed to determine posture quality.

- Elbow Angle Rule: Ensures proper extension and contraction.
- Back Posture Rule: Prevents excessive leaning.
- Arm Elevation Rule: Arms must reach shoulder height in lateral raises.
- Wrist Alignment Rule: Maintains wrist-shoulder alignment.
- Symmetry Rule: Left and right arms should move symmetrically.

8. Rep Counting Mechanism

Repetition counting is implemented using a finite-state machine. A repetition is counted only after detecting a complete and valid movement cycle.

9. Real-Time Feedback

The system provides instant feedback such as 'Extend your arm fully', 'Keep your back straight', and 'Raise arms to shoulder level' to help users correct their posture while exercising.

10. User Interface

The application is built using Streamlit and includes controls for starting and stopping the webcam, uploading videos, resetting repetition counters, and visualizing live pose landmarks.

11. Conclusion

This project demonstrates a lightweight and explainable approach to exercise form analysis using pose estimation. By relying on rule-based logic, the system provides reliable feedback without requiring machine learning model training.